tf.reshape Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/reshape

> *Reshapes a tensor.*

Function Signatures

```
tf.reshape( tensor, shape, name=None )
```

Code Examples

```
tf.reshape(
tensor, shape, name=None
)
```

```
tf.reshape(
tensor, shape, name=None
)
```

```
t1 = [[1, 2, 3],
[4, 5, 6]]
print(tf.shape(t1).numpy())
[2 3]
t2 = tf.reshape(t1, [6])
t2

tf.reshape(t2, [3, 2])
```

```
t1 = [[1, 2, 3],
```

```
print(tf.shape(t1).numpy())
```

```
t2 = tf.reshape(t1, [6])
```

Documentation

Reshapes a tensor.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.reshape

Used in the notebooks

• Introduction to Tensors

- Advanced automatic differentiation
- DTensor concepts
- Multilayer perceptrons for digit recognition with Core APIs
- Distributed training with Core APIs and DTensor
- Learned data compression
- Adversarial example using FGSM
- Convolutional Variational Autoencoder
- Load CSV data
- Load text

Given tensor, this operation returns a new tf.Tensor that has the same values as tensor in the same order, except with a new shape given by shape.

The tf.reshape does not change the order of or the total number of elements in the tensor, and so it can reuse the underlying data buffer. This makes it a fast operation independent of how big of a tensor it is operating on.

To instead reorder the data to rearrange the dimensions of a tensor, see tf.transpose.

If one component of shape is the special value -1, the size of that dimension is computed so that the total size remains constant. In particular, a shape of [-1] flattens into 1-D. At most one component of shape can be -1.

tf.reshape(t, []) reshapes a tensor t with one element to a scalar.

More examples:

Returns